

JEE Mains 2019 Chapter wise Question Bank

Motion in One Dimension - Questions

Q1

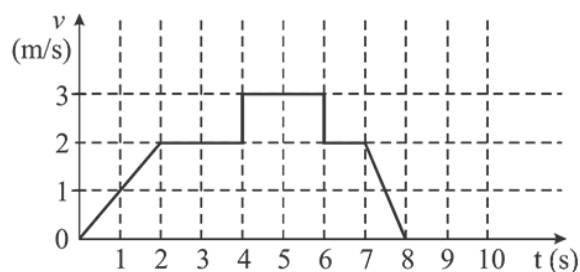
In a car race on straight road, car A takes a time t less than car B at the finish and passes finishing point with a speed ' v ' more than of car B. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then ' v ' is equal to:

- (1) $\frac{2a_1 a_2}{a_1 + a_2} t$ (2) $\sqrt{2a_1 a_2} t$
 (3) $\sqrt{a_1 a_2} t$ (4) $\frac{a_1 + a_2}{2} t$

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Q2

A particle starts from the origin at time $t = 0$ and moves along the positive x -axis. The graph of velocity with respect to time is shown in figure. What is the position of the particle at time $t = 5$ s?



- (1) 10 m (2) 6 m
 (3) 3 m (4) 9 m

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Q3

A passenger train of length 60 m travels at a speed of 80 km/hr. Another freight train of length 120 m travels at a speed of 30 km/h. The ratio of times taken by the passenger train to completely cross the freight train when: (i) they are moving in same direction, and (ii) in the opposite directions is:

- (1) $\frac{11}{5}$ (2) $\frac{5}{2}$
 (3) $\frac{3}{2}$ (4) $\frac{25}{11}$

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Q4

Ship A is sailing towards north-east with velocity $\vec{v} = 30\hat{i} + 50\hat{j}$ km/hr where $\hat{i}$ points east and $\hat{j}$, north. Ship B is at a distance of 80 km east and 150 km north of Ship A and is sailing towards west at 10 km/hr. A will be at minimum distance from B in:

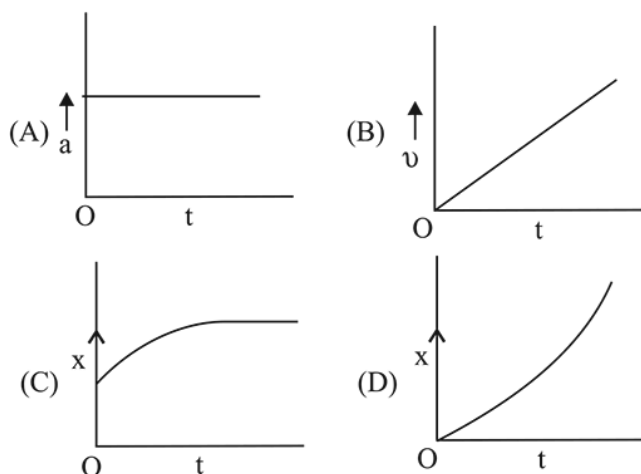
- (1) 4.2 hrs (2) 2.6 hrs (3) 3.2 hrs (4) 2.2 hrs

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Q5

Motion in One Dimension

A particle starts from origin O from rest and moves with a uniform acceleration along the positive x -axis. Identify all figures that correctly represents the motion qualitatively (a = acceleration, v = velocity, x = displacement, t = time)

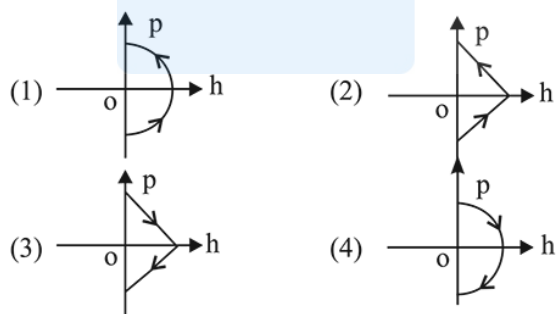


- (1) (B), (C) (2) (A)
(3) (A), (B), (C) (4) (A), (B), (D)

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Q6

A ball is thrown vertically up (taken as + z -axis) from the ground. The correct momentum-height (p - h) diagram is:



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Q7

The position of a particle as a function of time t , is given by $x(t) = at + bt^2 - ct^3$

where, a , b and c are constants. When the particle attains zero acceleration, then its velocity will be:

- (1) $a + \frac{b^2}{4c}$ (2) $a + \frac{b^2}{3c}$
(3) $a + \frac{b^2}{c}$ (4) $a + \frac{b^2}{2c}$

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Q8

A man (mass = 50 kg) and his son (mass = 20 kg) are standing on a frictionless surface facing each other. The man pushes his son so that he starts moving at a speed of 0.70 ms^{-1} with respect to the man. The speed of the man with respect to the surface is :

- (1) 0.28 ms^{-1} (2) 0.20 ms^{-1}
(3) 0.47 ms^{-1} (4) 0.14 ms^{-1}

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Q9

A particle is moving with speed $v = b\sqrt{x}$ along positive x -axis. Calculate the speed of the particle at time $t = \tau$ (assume that the particle is at origin at $t = 0$).

- (1) $\frac{b^2\tau}{4}$ (2) $\frac{b^2\tau}{2}$ (3) $b^2\tau$ (4) $\frac{b^2\tau}{\sqrt{2}}$

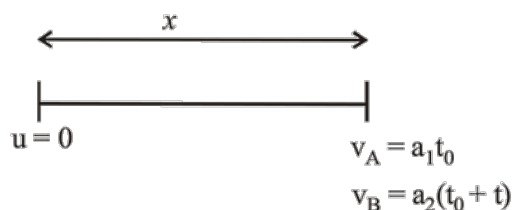
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Motion in One Dimension - Answers

Q1

- (3) Let time taken by A to reach finishing point is t_0
 $\therefore$ Time taken by B to reach finishing point = $t_0 + t$



$$v_A - v_B = v$$

$$\Rightarrow v = a_1 t_0 - a_2 (t_0 + t) = (a_1 - a_2)t_0 - a_2 t \quad \dots(i)$$

$$x_B = x_A = \frac{1}{2} a_1 t_0^2 = \frac{1}{2} a_2 (t_0 + t)^2$$

$$\Rightarrow \sqrt{a_1} t_0 = \sqrt{a_2} (t_0 + t)$$

$$\Rightarrow (\sqrt{a_1} - \sqrt{a_2}) t_0 = \sqrt{a_2} t$$

$$\Rightarrow t_0 = \frac{\sqrt{a_2} t}{\sqrt{a_1} - \sqrt{a_2}}$$

Putting this value of t_0 in equation (i)

$$v = (a_1 - a_2) \frac{\sqrt{a_2} t}{\sqrt{a_1} - \sqrt{a_2}} - a_2 t$$

$$= (\sqrt{a_1} + \sqrt{a_2}) \sqrt{a_2} t - a_2 t$$

$$= \sqrt{a_1 a_2} t + a_2 t - a_2 t$$

$$\text{or, } v = \sqrt{a_1 a_2} t$$

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Q2

- (4) Position of the particle,
 $S = \text{area under graph (time } t = 0 \text{ to } 5\text{s})$

$$= \frac{1}{2} \times 2 \times 2 + 2 \times 2 + 3 \times 1 = 9\text{m}$$

10 Jan Evening

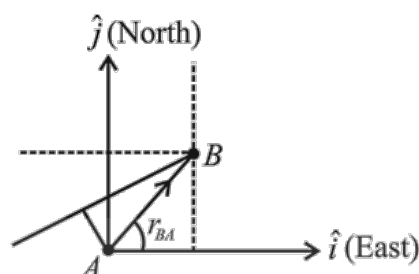
Q3

(1)

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Q4

(2)



$$\vec{v}_A = 30\hat{i} + 50\hat{j} \text{ km/hr}$$

$$\vec{v}_B = (-10\hat{i}) \text{ km/hr}$$

$$\vec{r}_{BA} = (80\hat{i} + 150\hat{j}) \text{ km}$$

$$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = -10\hat{i} - 30\hat{i} - 50\hat{j}$$

$$= 40\hat{i} - 50\hat{j}$$

$$t_{\text{minimum}} = \frac{|\vec{r}_{BA} \cdot \vec{v}_{BA}|}{|\vec{v}_{BA}|^2}$$

$$= \frac{|(80\hat{i} + 150\hat{j}) \cdot (-40\hat{i} - 50\hat{j})|}{(10\sqrt{41})^2}$$

$$\therefore t = \frac{10700}{10\sqrt{41} \times 10\sqrt{41}} = \frac{107}{41} = 2.6 \text{ hrs.}$$

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Q5

- (4) For constant acceleration, there is straight line parallel to t-axis on $\vec{a}-t$.

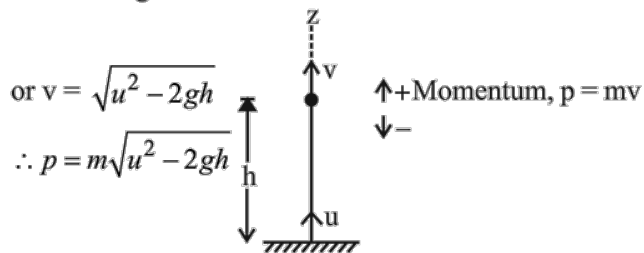
Inclined straight line on $\vec{v}-t$, and parabola on $\vec{x}-t$.

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Motion in One Dimension

Q6

$$(4) v^2 = u^2 - 2gh$$



Therefore graph between p and h cannot have straight line.

(2) and (3) are not possible.

During upward journey as h increases, p decreases and in downward journey as h decreases p increases. Therefore 4 is the correct option.

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Q7

$$(2) x = at + bt^2 - ct^3$$

$$\text{Velocity, } v = \frac{dx}{dt} = \frac{d}{dt}(at + bt^2 + ct^3)$$

$$= a + 2bt - 3ct^2$$

$$\text{Acceleration, } \frac{dv}{dt} = \frac{d}{dt}(a + 2bt - 3ct^2)$$

$$\text{or } 0 = 2b - 3c \times 2t$$

$$\therefore t = \left(\frac{b}{3c}\right)$$

$$\text{and } v = a + 2b\left(\frac{b}{3c}\right) - 3c\left(\frac{b}{3c}\right)^2$$

$$= a + \frac{b^2}{3c}$$

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Q8

$$(2) P_i = P_f$$

$$\text{or } 0 = 20(0.7 - v) = 50v$$

$$\text{or } v = 0.2 \text{ m/s}$$

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Q9

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$$(2) B_0 = \frac{E_0}{C} = \frac{60}{3 \times 10^8}$$

$$= 20 \times 10^{-8} \text{ T} = 2 \times 10^{-7} \text{ T}$$

$$K = \frac{\omega}{v} = \frac{2\pi f}{v} = \frac{2\pi \times 23.9 \times 10^9}{3 \times 10^8} = 500$$

$$\text{Therefore, } \vec{B} = B_0 \sin(kz - \omega t)$$

$$= 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$$

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